

Anugrah Surya Putra

Mobile Engineer

South Jakarta, Jakarta | anugrahputra@gmail.com | downormal.dev | linkedin.com/in/anugrahputra
github.com/anugrahputra

Summary

Mobile Engineer with 2+ years of experience in **Flutter** and **Kotlin Multiplatform**. Currently the sole mobile engineer at BRIK, building offline-first apps for field operations teams and owning everything from module architecture to **CI/CD** and over-the-air releases.

Education

Universitas Pasundan, Bachelor of Engineering (BE) in Informatics Engineering – Kota Bandung, Sept 2018 – June 2023
Jawa Barat

Experience

Mobile Engineer, PT Bangun Rancang Indonesia Kita (BRIK) – South Jakarta, Jakarta Aug 2025 – present

- Sole mobile engineer for BRIK's internal **Flutter** apps, covering six field roles across logistics operations. Own architecture, delivery, and production support with no other mobile engineer on the team.
- Leading a phased rewrite from page-based code to **feature-first modules**, each role isolated behind its own **Bloc** layer, **Drift** store, and route table. Shipped the first modules to production across two major releases, with the rest moving over one at a time rather than in a single cutover.
- Rebuilt a core operations flow as an offline-first module. Wrote the **WebSockets** client for on-site hardware with per-request reply matching, timeouts, and token refresh on reconnect, falling back to REST-only manual entry when the link drops. Edits send only changed fields, and offline writes reach the server in order once back online. Covered by 300+ module tests.
- Designed the **offline-first sync layer** around **Drift** for local persistence and an **outbox queue** that replays writes once connectivity returns. Built background **GPS tracking** on a foreground service with its own retry queue, hardened against OEM battery killers on MIUI devices.
- Own the release pipeline end to end. **Bitbucket Pipelines** for build and test with tests gating every build, **Google Play Console** for store releases, **Shorebird** for over-the-air patches. Shipped 20+ store releases and 30+ OTA patches, cutting hotfix delivery from days to same-day without store review.
- Cut startup and runtime cost by removing double JSON parsing in the response layer, adding **HTTP/2** connection pooling, and deferring non-critical init off the launch path. Moved photo and video processing into **FFmpeg** background tasks to stop camera stalls and ANRs on low-end devices.
- Instrumented the app with **Sentry**, added **Maestro** E2E coverage on critical user flows, and work directly with field users to turn their reports into fixes.

Mobile Engineer, PT. Semesta Arus Technology – South Jakarta, Jakarta June 2024 – July 2025

- Developed cross-platform mobile applications using **Flutter** and **Compose Multiplatform**, delivering unified experiences for iOS and Android.
- Implemented **Clean Architecture** and **SOLID** principles, creating modular codebases that improved testability and reduced feature development time.
- Partnered with backend teams to integrate **RESTful APIs**, ensuring reliable data synchronization for business-critical operations.
- Profiled and resolved memory leaks and performance bottlenecks, reducing app crash rates by 15% and improving overall stability.
- Set up automated build and testing pipelines, reducing deployment preparation time and ensuring consistent release quality.
- Mentored junior developers on **Flutter** best practices and code review processes, improving team code quality and knowledge sharing.

Projects

Quran API

Oct 2025 – Mar 2026

Go-based REST API for retrieving Quranic texts, Tafsir (exegesis), and prayer times, featuring full-text search capabilities using **Bleve** and **Redis** caching.

- Architected a **Clean Architecture** backend in **Go** using the **Gin** framework, decoupling handlers, business services, and repository layers to ensure modularity.
- Engineered a high-performance full-text search engine using **Bleve**, configuring a custom N-gram text analyzer and Unicode tokenizer to enable fuzzy matches on transliterated queries.
- Optimized API response latency by integrating **Redis** as a cache layer with a 24-hour TTL for external upstream APIs, reducing downstream request overhead.
- Developed a custom IP-based fixed-window rate limiter middleware with **Redis**, utilizing a fail-open design to maintain API availability during database or cache disruptions.
- Streamlined container deployments by building a multi-stage **Docker** pipeline that pre-builds and bundles search indexes to eliminate cold-start initialization latency.

Portfolio Backend

Mar 2026 – present

Go-based REST API backend for a personal portfolio system using **Clean Architecture**, **Gin**, **PostgreSQL**, and **SQLC**.

- Architected a modular system using Clean (Hexagonal) Architecture to decouple business logic, database operations, and the HTTP presentation layer.
- Designed a high-performance LATERAL JOIN query using **PostgreSQL** JSON aggregation (`jsonb_agg`) to retrieve complex, nested resume hierarchies in a single database roundtrip.
- Utilized **SQLC** for type-safe database queries and schema-first code generation, reducing boilerplate code and improving performance with `pgx/v5` connection pooling.
- Developed a pointer-based fetch-and-merge strategy to safely handle partial updates (PATCH/PUT) of domain resources without overwriting unset fields.
- Implemented table-driven unit tests using Testify and mock interfaces to ensure high-coverage validation of use cases and HTTP endpoints.

Change Project Name

Sept 2025 – present

Open-source CLI tool built with **Dart** to automate **Flutter/Dart** project renaming and refactoring.

- Built a recursive file-traversal engine that safely refactors imports and configurations while skipping build artifacts.
- Implemented RegEx validation with a dry-run mode for predictable, risk-free project transformations.
- Automated post-rename toolchain execution (`flutter clean`, `pub get`) for immediate project consistency.

Cosmic App KIOSK Touchscreen

Oct 2024 – Feb 2025

Interactive KIOSK information display app for DPR RI built with **Kotlin** and **Jetpack Compose**.

- Developed automated power-scheduling and WebView reset mechanisms for reliable unattended operation.
- Built real-time bidirectional communication using **WebSockets** with heartbeat monitoring for remote status tracking.
- Designed intuitive touch-based UI optimized for large-screen KIOSK displays with accessibility considerations.
- Implemented offline-first data caching to ensure continuous operation during network interruptions.

E-Market Mobile App

Dec 2022 – May 2023

Dual-sided e-commerce platform (Seller & Buyer) for regional MSMEs built with **Flutter** and **Firebase**.

- Developed full-stack mobile experience with **Flutter**, **Firebase**, and **Google Maps API** for location-based services.
- Conducted rigorous blackbox testing, achieving 95%+ test pass rate before production deployment.
- Integrated **Firebase** Authentication and Cloud Firestore for real-time inventory management and order tracking.
- Designed responsive UI components supporting both seller and buyer workflows within a unified codebase.

Skills

Languages: Dart, Kotlin, Go, Java, Swift

Frameworks: Flutter, Compose Multiplatform, Jetpack Compose, SwiftUI

Libraries: Bloc, Freezed, Drift, Hive, Dio, GoRouter, `get_it`, `injectable`

Tools & Platforms: Firebase, Sentry, Shorebird, Bitbucket Pipelines, Maestro, Git, Figma, Android Studio, VS Code, Neovim

Practices: Clean Architecture, SOLID, CI/CD, TDD, Agile Methodology